MAGIC SQUARES OF SQUARES AND ELLIPTIC DIVISIBILITY SEQUENCES: THE RANK-ONE CASE

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ABSTRACT. Robertson observed that a 3×3 magic square of squares gives three points on a congruent number curve $E_n\colon y^2 = x^3 - n^2x$ whose x-coordinates are in arithmetic progression, and three more such points on a second congruent number curve. Bremner, Silverman and Tzanakis proved that for squarefree n a subgroup of $E_n(\mathbb{Q})$ of rank one contains no non-trivial *integral* arithmetic progression. Their theorem never applies to magic squares of squares: in a primitive one the common differences are divisible by 8, and as a result the points never have integral coordinates. We prove a rational analogue on the curves $y^2 = x^3 + Dx$: the x-coordinates of three multiples of a non-torsion point with square x-coordinate, with pairwise distinct absolute indices, are never in arithmetic progression. The proof uses the primitive divisor theorem of Voutier and Yabuta, with Verzobio's correction. Consequently, in any magic square of squares each of the two families of three points generates a subgroup of rank at least two. Equivalently, the common difference of either diagonal of the square has a squarefree part n with $\operatorname{rank} E_n(\mathbb{Q}) \geq 2$. By a theorem of Smith, such n form a set of density zero.

1. INTRODUCTION

A *magic square of squares* is a 3×3 array of nine distinct positive integer squares whose rows, columns and two diagonals all have the same sum. Whether one exists is a well-known open problem, popularized by LaBar [11] and by Gardner (see Boyer [2] for the history, the prizes, and near-misses such as Bremner's magic square with seven square entries). Robertson [15] related the problem to congruent number curves, and Bremner [3, 4] studied it through its sub-configurations and their geometry. Bruin, Thomas and Várilly-Alvarado [6] proved that the associated surface is algebraically quasi-hyperbolic. Auel and Singer [1] have recently described its algebraic geometry in detail, and Kominers [10] has studied magic squares of squares with center 0 over number fields. Extensive computer searches have found no example.

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Every 3×3 magic square of integers can be written in the Lucas form

$$\begin{pmatrix} E+a & E-a-b & E+b \\ E-a+b & E & E+a-b \\ E-b & E+a+b & E-a \end{pmatrix}, \qquad E, a, b \in \mathbb{Z}, \tag{1}$$

whose entries are the nine numbers $L_{\alpha,\beta} = E + \alpha a + \beta b$ with $\alpha, \beta \in \{-1, 0, 1\}$. The entries are distinct exactly when $ab \neq 0$, $a \neq \pm b$, $a \neq \pm 2b$ and $b \neq \pm 2a$. The main diagonal is $(E-a, E, E+a)$ and the anti-diagonal is $(E-b, E, E+b)$.

For fixed α, the three entries $L_{\alpha,-1}, L_{\alpha,0}, L_{\alpha,1}$ form an arithmetic progression with common difference b. In (1) they are the anti-diagonal ($\alpha = 0$) and the two broken diagonals parallel to it. We call these the three *b-progressions*; they partition the nine entries. Likewise, for fixed β, the entries $L_{-1,\beta}, L_{0,\beta}, L_{1,\beta}$ form the three *a-progressions*, with common difference a: the main diagonal and its two broken parallels.

For a nonzero integer m, write $\mathrm{sqf}(m)$ for the squarefree part of $|m|$: the squarefree integer $n \geq 1$ with $|m| = nk^2$ for some integer k. For a squarefree $n \geq 1$, let E_n be the congruent number curve

$$E_n \colon \; y^2 = x^3 - n^2 x.$$

If all nine entries of (1) are squares, write $|b| = nk^2$ with $n = \mathrm{sqf}(b)$. For each α, the three entries of the α-th b-progression divided by k^2 are the rational squares $x_\alpha - n$, x_α and $x_\alpha + n$, where $x_\alpha = (E + \alpha a)/k^2$. So $x_\alpha(x_\alpha - n)(x_\alpha + n)$ is a rational square, and there is a point

$$P_\alpha = (x_\alpha, y_\alpha) \in E_n(\mathbb{Q}), \qquad x_{-1} + x_1 = 2x_0. \tag{2}$$

This is the observation of Robertson [15]. The a-progressions give three points on $E_{\mathrm{sqf}(a)}$ in the same way.

Motivated by this connection, Bremner, Silverman and Tzanakis proved in [5] that if n is squarefree then a subgroup of $E_n(\mathbb{Q})$ of rank one contains no non-trivial *integral* arithmetic progression. Their proof uses delicate height estimates. Their theorem does not reach magic squares of squares. In a primitive magic square of squares all nine entries are odd and 8 divides a and b (Remark 5.1), so b is never squarefree, and the points (2) never have integral coordinates. Our main result is a rational version, for the points that magic squares actually produce.

Theorem 1.1. *Let E, a, b be integers with $ab \neq 0$ such that all nine entries of (1) are squares of integers. Then the points P_{-1}, P_0, P_1 of (2) generate a subgroup of $E_{\mathrm{sqf}(b)}(\mathbb{Q})$ of rank at least two. Likewise, the three points coming from the a-progressions generate a subgroup of $E_{\mathrm{sqf}(a)}(\mathbb{Q})$ of rank at least two.*

Corollary 1.2. *Let d be the common difference of either diagonal of a magic square of squares. Then $n = \mathrm{sqf}(d)$ is a congruent number with $\operatorname{rank} E_n(\mathbb{Q}) \geq 2$. The set of such n has natural density zero among squarefree integers. Its elements up to 150 are $34, 41, 65, 137, 138, 145$.*

Example 1.3. The curve E_6 has rank one; its point $2G$, for the generator $G = (-3, 9)$, has $x(2G) = 25/4$, which corresponds to the progression $1, 25, 49$ of difference $24 = 6 \cdot 2^2$. By Corollary 1.2, no magic square of squares has a diagonal whose common difference is of the form $6k^2$. In particular, $(1, 25, 49)$ is not a diagonal of any magic square of squares.

The key input is Lemma 3.1. Let Q be a non-torsion point on an elliptic curve with a Weierstrass equation with integer coefficients, and suppose that every term of the elliptic divisibility sequence of Q from the third on has a primitive divisor. Then the x-coordinates of three multiples of Q with pairwise distinct absolute indices are never in arithmetic progression. Its proof is short. On the curves $y^2 = x^3 + Dx$ with D fourth-power-free in minimal form, the primitive divisor theorem of Voutier and Yabuta [23], which appeared after [5], supplies the hypothesis whenever $x(Q)$ is a square (Theorem 2.2). More general results on arithmetic progressions among x-coordinates of rational points, such as Choi's additive rigidity theorem [7], bound the number of points in such progressions in terms of the Mordell–Weil rank, but do not give this exclusion. As far as we know, Lemma 3.1 and Theorem 1.1 are new. Both, and the rank statement of Corollary 1.2, have been checked in the Lean proof assistant, with Theorem 2.2 and the torsion of $E_n(\mathbb{Q})$ as explicit hypotheses (Remark 5.6).

2. Congruent number curves and their divisibility sequences

Lemma 2.1. *Let $n \geq 1$ be squarefree.*

(a) $E_n(\mathbb{Q})_{\mathrm{tors}} = E_n[2] = \{O, (0,0), (n,0), (-n,0)\}$.
(b) *A point $P = (x, y) \in E_n(\mathbb{Q})$ lies in $2E_n(\mathbb{Q})$ if and only if x, $x - n$ and $x + n$ are all squares in $\mathbb{Q}$.*
(c) $2E_n(\mathbb{Q})$ *is torsion-free.*
(d) *The equation $y^2 = x^3 - n^2x$ is a global minimal model of E_n.*

Proof. Part (a) is classical; see [9, Ch. I, §9]. Part (b) is the standard criterion for halving a point on $y^2 = (x - e_1)(x - e_2)(x - e_3)$ with $e_i \in \mathbb{Q}$, applied with $\{e_i\} = \{0, n, -n\}$; see [8, Ch. IV]. For (c), if $T = 2R \in 2E_n(\mathbb{Q})$ is torsion then so is R, so $R \in E_n[2]$ by (a), and $T = O$.

For (d), the discriminant is $\Delta = 64n^6$. At an odd prime p, $v_p(\Delta) = 6v_p(n) \leq 6 < 12$, so the model is minimal at p. If n is odd, then $v_2(\Delta) = 6$ and the same holds at 2. If $n = 2m$ with m odd, then replacing m by mu^2 with $u \in \mathbb{Z}_2^\times$ gives an isomorphic model over $\mathbb{Z}_2$, via $(x, y) \mapsto (u^2x, u^3y)$. So minimality at 2 depends only on m modulo $(\mathbb{Z}_2^\times)^2$, that is, on m mod 8. Tate's algorithm (we used `ellminimalmodel` in PARI/GP [14]) shows that the model is minimal at 2 for $n = 2, 6, 10, 14$, which covers $m \equiv 1, 3, 5, 7 \pmod 8$. □

Now let C be an elliptic curve over $\mathbb{Q}$ given by a Weierstrass equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \qquad a_1, a_2, a_3, a_4, a_6 \in \mathbb{Z}, \tag{3}$$

and for a prime ℓ let v_ℓ be the ℓ-adic valuation on $\mathbb{Q}$. If $(x, y) \in C(\mathbb{Q})$ and $t = -v_\ell(x) > 0$, then t is even. Indeed, the right-hand side of (3) has valuation $-3t$, because x^3 does and the other terms have valuation at least $-2t$. If $v_\ell(y) \geq -t$, every term on the left has valuation at least $-2t$, which is impossible. So $v_\ell(y) < -t$. Then y^2 has strictly smaller valuation than a_1xy and a_3y, and comparing the two sides gives $2v_\ell(y) = -3t$. Consequently, for $P \in C(\mathbb{Q}) \setminus \{O\}$ we can write $x(P) = A(P)/B(P)^2$ with $A(P), B(P) \in \mathbb{Z}$ coprime and $B(P) \geq 1$, and for every prime ℓ

$$\ell \mid B(P) \iff v_\ell\big(x(P)\big) < 0 \iff v_\ell\big(x(P)\big) = -2v_\ell\big(B(P)\big) \leq -2. \tag{4}$$

Since $-P$ has the same x-coordinate as P, $B(-P) = B(P)$. For a non-torsion point Q, the integers $B_m = B(mQ)$, $m \geq 1$, form the *elliptic divisibility sequence* of Q [24]. A prime ℓ is a *primitive divisor* of B_M if $\ell \mid B_M$ and $\ell \nmid B_m$ for $1 \leq m < M$. Silverman [16] showed that all but finitely many terms have a primitive divisor. For the curves $W_D\colon y^2 = x^3 + Dx$, where $D \neq 0$ is an integer, there is a uniform statement. Note that $E_n = W_{-n^2}$.

Theorem 2.2 (Voutier–Yabuta [23], Verzobio [22])**.** *Let D be a fourth-power-free integer for which $y^2 = x^3 + Dx$ is a minimal model, and let Q be a non-torsion rational point on it with $x(Q)$ a square in $\mathbb{Q}$. Then $B_M(Q)$ has a primitive divisor for every $M \geq 3$.*

Voutier and Yabuta [23] prove that B_M has a primitive divisor for every even $M > 2$, and for every odd $M > 1$ when $x(Q) < 0$ or $x(Q)$ is a square. Their proof needs the case $M = 5m$ (their Lemma 5.1). Verzobio [22] found a gap in the proof of that lemma and proved, for minimal models $y^2 = x^3 + Dx$, that B_{5m} has a primitive divisor for every m.

3. Multiples of one point are never in arithmetic progression

Lemma 3.1. *Let C be an elliptic curve over $\mathbb{Q}$ given by a Weierstrass equation* (3) *with integer coefficients, and let $Q \in C(\mathbb{Q})$ be a non-torsion point such that $B_M(Q)$ has a primitive divisor for every $M \geq 3$. Let m_{-1}, m_0, m_1 be nonzero integers whose absolute values are pairwise distinct. Then*

$$x(m_{-1}Q) + x(m_1Q) \neq 2\,x(m_0Q).$$

By Theorem 2.2, the hypothesis on Q holds when C is $y^2 = x^3 + Dx$ with D fourth-power-free, the model is minimal, and $x(Q)$ is a square in $\mathbb{Q}$.

Proof. The three integers $|m_\alpha|$ are distinct and positive, so their maximum M satisfies $M \geq 3$. Let ℓ be a primitive divisor of B_M, and let α^* be the index with $|m_{\alpha^*}| = M$. Since $x(mQ) = x(|m|Q)$, (4) gives

$$v_\ell\big(x(m_{\alpha^*}Q)\big) = -2v_\ell(B_M) \leq -2, \qquad v_\ell\big(x(m_\beta Q)\big) \geq 0 \quad (\beta \neq \alpha^*),$$

because $\ell \nmid B_{|m_\beta|}$ when $|m_\beta| < M$.

If $\alpha^* = \pm 1$, the left-hand side $x(m_{-1}Q) + x(m_1Q)$ has ℓ-adic valuation at most -2. The right-hand side $2x(m_0Q)$ has valuation at least 0.

If $\alpha^* = 0$, the right-hand side has valuation $v_\ell(2) - 2v_\ell(B_M) \le 1 - 2 < 0$. The left-hand side has valuation at least 0.

In either case the two sides are different. □

4. Proof of Theorem 1.1 and Corollary 1.2

Proof of Theorem 1.1. Let $n = \mathrm{sqf}(b)$ and let $P_{-1}, P_0, P_1 \in E_n(\mathbb{Q})$ be the points of (2).

- By Lemma 2.1(b), each P_α lies in $2E_n(\mathbb{Q})$, since x_α and $x_\alpha \pm n$ are squares.
- The $x_\alpha = (E + \alpha a)/k^2$ are pairwise distinct because $a \neq 0$, and none of the P_α is O.
- Let $\Gamma = \langle P_{-1}, P_0, P_1 \rangle \subseteq 2E_n(\mathbb{Q})$. By Lemma 2.1(c), Γ is a nonzero, torsion-free, finitely generated abelian group, so $\operatorname{rank}\Gamma \ge 1$.

Suppose $\operatorname{rank}\Gamma = 1$. Then Γ is infinite cyclic, $\Gamma = \langle Q \rangle$, with $Q \in \Gamma \subseteq 2E_n(\mathbb{Q})$ non-torsion, so $x(Q)$ is a square by Lemma 2.1(b). Write $P_\alpha = m_\alpha Q$ with $m_\alpha \in \mathbb{Z} \setminus \{0\}$. For a non-torsion Q, $x(mQ) = x(m'Q)$ implies $mQ = \pm m'Q$, and hence $m = \pm m'$. So the distinctness of the x_α forces the $|m_\alpha|$ to be pairwise distinct. Since $D = -n^2$ is fourth-power-free and the model is minimal (Lemma 2.1(d)), Theorem 2.2 shows that $B_M(Q)$ has a primitive divisor for every $M \ge 3$. So Lemma 3.1 applies, and it contradicts $x_{-1} + x_1 = 2x_0$. Hence $\operatorname{rank}\Gamma \ge 2$.

The statement for the a-progressions follows by exchanging the roles of a and b, which needs $b \neq 0$. □

Proof of Corollary 1.2. The common differences of the two diagonals of (1) are a and b, and $ab \neq 0$ for a magic square of squares. By Theorem 1.1, $\operatorname{rank} E_n(\mathbb{Q}) \ge 2$ for $n = \mathrm{sqf}(a)$ and $n = \mathrm{sqf}(b)$; in particular n is a congruent number.

For the density, E_n is the quadratic twist by n of $E_1 \colon y^2 = x^3 - x$. The curve E_1 has full rational 2-torsion. It has no rational cyclic subgroup of order four, because the x-coordinates of its points of exact order four are the roots of $x^2 + 1$ and $x^2 \pm 2x - 1$, none of which is rational. Smith [19] proved that for such a curve the number of twists E_1^d with $|d| < N$ and rank at least two is $o(N)$.

For the list, PARI/GP's `ellrank` [14] proves $\operatorname{rank} E_n(\mathbb{Q}) \ge 2$ for $n = 34, 41, 65, 137, 138, 145$, and $\operatorname{rank} E_n(\mathbb{Q}) \le 1$ for every other squarefree $n \le 150$ except $n = 113$, where it gives $0 \le \text{rank} \le 2$. For $n = 113$, Tunnell's criterion [21] settles the question. If an odd squarefree n is congruent then $\#\{x^2 + 2y^2 + 8z^2 = n\} = 2\,\#\{x^2 + 2y^2 + 32z^2 = n\}$. For $n = 113$ these counts are 32 and 24, so 113 is not congruent, and $\operatorname{rank} E_{113}(\mathbb{Q}) = 0$. □

5. Remarks

Remark 5.1 (Relation to [5]). Every magic square of squares is a square multiple of a primitive one, whose entries have no common factor greater

than 1, and the property in Theorem 1.1 is unchanged by scaling. In a primitive one, write the center as $E = e^2$. Each line through the center is $(e^2 - t, e^2, e^2 + t)$ with $t \in \{a, b, a + b, a - b\}$, and $e^2 - t = x^2$, $e^2 + t = y^2$ give $x^2 + y^2 = 2e^2$.

- If e were even, then $x^2 + y^2 \equiv 0 \pmod 4$ would force x and y to be even for every such line, making all nine entries even. So e is odd.
- Then $x^2 + y^2 = 2e^2 \equiv 2 \pmod 4$ forces x and y odd, so all nine entries are odd and $t = y^2 - e^2 \equiv 0 \pmod 8$. In particular $8 \mid a$ and $8 \mid b$.

Writing $|b| = nk^2$ with n squarefree, $v_2(n) \leq 1$ forces k to be even. So $x_\alpha = (E + \alpha a)/k^2$ has an odd numerator and an even denominator, and is never an integer. Theorem 1.1 of [5], which concerns integral progressions on E_n with n squarefree, therefore never applies to the points (2); nor does their Theorem 1.4 on large squarefree twists. Conversely, Lemma 3.1 and Theorem 2.2 do not recover [5, Thm. 1.1]. That theorem allows torsion points, and points outside $2E_n(\mathbb{Q})$, to which Theorem 2.2 need not apply.

Remark 5.2 (What the argument uses). The proof uses that all three b-progressions (or all three a-progressions) consist of squares. Since these partition the nine entries, it uses all nine square conditions. It says nothing about magic squares with seven or eight square entries. For instance, Bremner's magic square with seven square entries [3] has $E = 425^2$, $a = -41496$ and $b = 138600$. Two of its b-progressions and one of its a-progressions consist of squares, and $\mathrm{sqf}(b) = 154$ while $\mathrm{sqf}(a) = 10374$. Here $\operatorname{rank} E_{154}(\mathbb{Q}) = 2$ and $\operatorname{rank} E_{10374}(\mathbb{Q}) = 1$, which is consistent with the theorem, since that square is not a magic square of squares.

Remark 5.3 (Near misses). Two full b-progressions already give two of the points (2), without the third b-progression and without a square center. These two points need not be independent, even in a magic square with nine distinct entries. For $E = 1201^2$, $a = 1319901$ and $b = 117600 = 6 \cdot 140^2$, the square (1) is

$$\begin{pmatrix} 2762302 & 70^2 & 1249^2 \\ 490^2 & 1201^2 & 2644702 \\ 1151^2 & 2879902 & 350^2 \end{pmatrix},$$

with magic sum 4327203 and six square entries. Its b-progressions with $\alpha = -1$ and $\alpha = 0$ are full, and the first of them is $70^2 \cdot (1, 25, 49)$. With $G = (-3, 9)$ and $H = 2G$ as in Example 1.3, we have $x_{-1} = 25/4 = x(H)$ and $x_0 = (1201/140)^2 = x(2H)$, so $P_{-1} = \pm H$ and $P_0 = \pm 2H$. Since $\operatorname{rank} E_6(\mathbb{Q}) = 1$, the hypothesis of Theorem 1.1 cannot be weakened to two full b-progressions.

In the near misses we have checked, however, the two points are independent. These are Bremner's square of Remark 5.2, whose b-progressions with $\alpha = 0, 1$ are full, and 2,320 configurations, found by a computer search over centers $E < 10^{12}$, in which the b-progressions with $\alpha = \pm 1$ are full and

E is not a square. The proof is exact. Write the two points as $P = 2R$ and $P' = 2R'$, as Lemma 2.1(b) allows. The map $(x, y) \mapsto (x, x - n)$, with $(0, 0) \mapsto (-1, -n)$ and $(n, 0) \mapsto (n, 2)$, induces an injective homomorphism $\phi\colon E_n(\mathbb{Q})/2E_n(\mathbb{Q}) \to (\mathbb{Q}^\times/\mathbb{Q}^{\times 2})^2$ [18, Prop. X.1.4]. In all but one case, ϕ maps R, R', $(0, 0)$ and $(n, 0)$ to elements that are linearly independent over $\mathbb{F}_2$. Then R and R' are independent modulo $E_n(\mathbb{Q})_{\mathrm{tors}}$, which is $E_n[2]$ by Lemma 2.1(a): a relation $cR + c'R' \in E_n[2]$ with c, c' coprime would give a nontrivial relation modulo $2E_n(\mathbb{Q})$. Hence P and P' are independent. The exception is a configuration with $n = 210$, in which ϕ maps R, R' and $(n, 0)$ to elements with product 1. Then $R + R' + (n, 0) = 2S$ for some $S \in E_n(\mathbb{Q})$, the same test succeeds for S and R', and $R \equiv 2S - R'$ modulo torsion. So in all these near misses $\operatorname{rank} E_{\mathrm{sqf}(b)}(\mathbb{Q}) \geq 2$, and the rank bound of Corollary 1.2 does not by itself separate magic squares of squares from them. The accompanying script `near_misses.gp` carries out these computations.

Remark 5.4 (Rank at least two). The valuation argument of Lemma 3.1 shows, in any rank, that a prime dividing the denominator of one of x_{-1}, x_0, x_1 must divide the denominator of another. When Γ has rank at least two, this becomes a condition on the common prime factors of the denominators of points in a lattice of rank at least two, that is, of the terms of an elliptic net [20]. Bounds of the required kind for the greatest common divisor of $B(P)$ and $B(Q)$, for independent P and Q, follow from Vojta's conjecture [17]. They are not known unconditionally, let alone uniformly in n. So we do not know how to extend Theorem 1.1 to rank two.

Remark 5.5 (Computations). As a check on Lemma 3.1, consider the 116 squarefree $n \leq 400$ for which PARI/GP, run with the random seed of the accompanying script `checks.gp`, proves $\operatorname{rank} E_n(\mathbb{Q}) = 1$ and finds a generator G, and set $H = 2G$. For ten more n, such as $n = 157$, it proves rank one but finds no generator. In each case, no three of $x(H), \ldots, x(16H)$ are in arithmetic progression, and every $B_m(H)$ with $2 \leq m \leq 16$ has a primitive divisor. The same script reproduces the other computations in this note, except those of Remark 5.3.

Remark 5.6 (Formal verification). Lemma 3.1, Theorem 1.1 and the rank statement of Corollary 1.2 are formalized in Lean 4 [13] with the Mathlib library [12], in the file `RankOne.lean` that accompanies this note. Two inputs enter as explicit hypotheses: the conclusion of Theorem 2.2 for E_n, and Lemma 2.1(a). Everything else is proved, including the halving formula behind Lemma 2.1(b), the group-law computations and the passage from a magic square of squares to the points (2).

- Lemma 3.1 is proved as stated, for every Weierstrass equation (3) with integer coefficients, together with the facts (4) about valuations that its proof uses.
- The rank of Γ is its rank as a free $\mathbb{Z}$-module.
- $\operatorname{rank} E_n(\mathbb{Q}) \geq 2$ is stated as the existence of two $\mathbb{Z}$-linearly independent points of $E_n(\mathbb{Q})$.

Lean reports that the proofs use only its three standard axioms. The density statement and the list in Corollary 1.2 are not formalized.

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